The Interaction of Financial Frictions and Uncertainty Shocks: Implications for the Cyclicality of Excess Returns

Giacomo Cattelan

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Motivating Facts

- Balance sheet of financial intermediaries is determining factor of macro dynamics

  $\Longrightarrow$ Higher weight on downside risk when balance sheet is deteriorated

- Increased volatility triggers **precautionary deleveraging**
  - $\downarrow$ investment in risky productive assets and hence $\downarrow$ output
  - $\uparrow$ of premium on risky asset and $\uparrow$ of excess returns

  $\Longrightarrow$ **Countercyclical excess returns**

- Distressed balance sheet $\Longrightarrow$ higher risk aversion $\Longrightarrow$ larger effect.

This Project

- **New evidence:** magnitude of countercyclical dynamics depends on

  uncertainty shocks $\times$ intermediaries balance sheet

- Differential responses to uncertainty shock conditional on the capitalization of the financial sector

- Macro-finance model with novel source of **financial uncertainty** about the tightness of the constraint

  $\Longrightarrow$ responses to increased uncertainty depends on the distance of the intermediaries from their financial constraint

- Global solution to detect state dependency of impulse responses.

Literature Review

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Specification

$$y_{t+h} = \alpha_h + \beta_0^h \epsilon_t^{UNC} + \beta_1^h \epsilon_t^{UNC} \eta_{t-1} + \boldsymbol{\gamma}' \cdot \mathbf{x}_t + u_t$$

$$y \in \{ \ (R_i^{exc})_{i=\text{Stocks, Corporate Bonds, Mortgages}},$$
$$C, I, \text{Hours worked}, Y,$$
$$\text{Credit to } NF, \text{ Treasuries, Net worth, Debt, Mortgages}\}$$ Data

$$\eta_{t-1} = \text{Y-o-Y } \%\Delta \frac{\text{Common Equity}_{t-1}}{\text{Invested Capital}_{t-1}}$$ Summary of η

- **Primary Dealers** for stocks and mortgages - He et al. (2017)
- **Financial Sector** for corporate bonds and other aggregates.

Specification

$$y_{t+h} = \alpha_h + \beta_0^h \epsilon_t^{UNC} + \beta_1^h \epsilon_t^{UNC} \eta_{t-1} + \boldsymbol{\gamma}' \cdot \mathbf{x}_t + u_t$$

- Uncertainty indexes: VXO as in Bloom (2009) and U_F^{LMN} as in Ludvigson et al. (2021) Details on LMN Indexes Plot
- Shocks identified as in the respective papers. Identification Shocks Plot

$$\widehat{IRF}(h \mid \eta_{-1} \geq 0) = \widehat{\beta_0^h} + \widehat{\beta_1^h}\overline{\eta} \qquad \overline{\eta} = \mathbb{E}[\eta_{-1} \mid \eta_{-1} \geq 0]$$

$$\widehat{IRF}(h \mid \eta_{-1} < 0) = \widehat{\beta_0^h} + \widehat{\beta_1^h}\underline{\eta} \qquad \underline{\eta} = \mathbb{E}[\eta_{-1} \mid \eta_{-1} < 0]$$ η

Excess Returns: shock to VXO

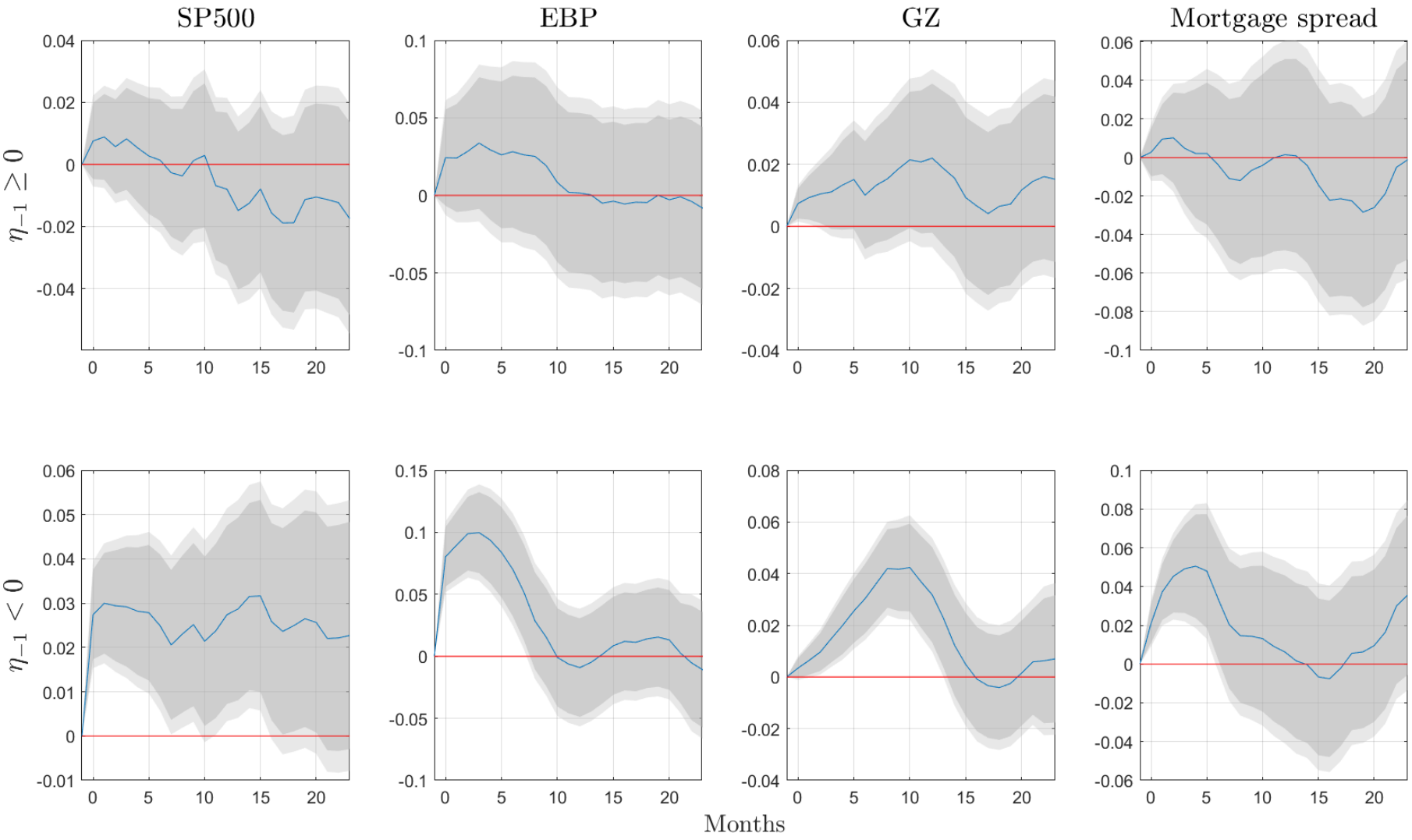

Figure: Dark grey: 90% confidence bands. Light grey: 95% confidence bands. Y-axis in percentage points. LMN

Real Variables: shock to VXO

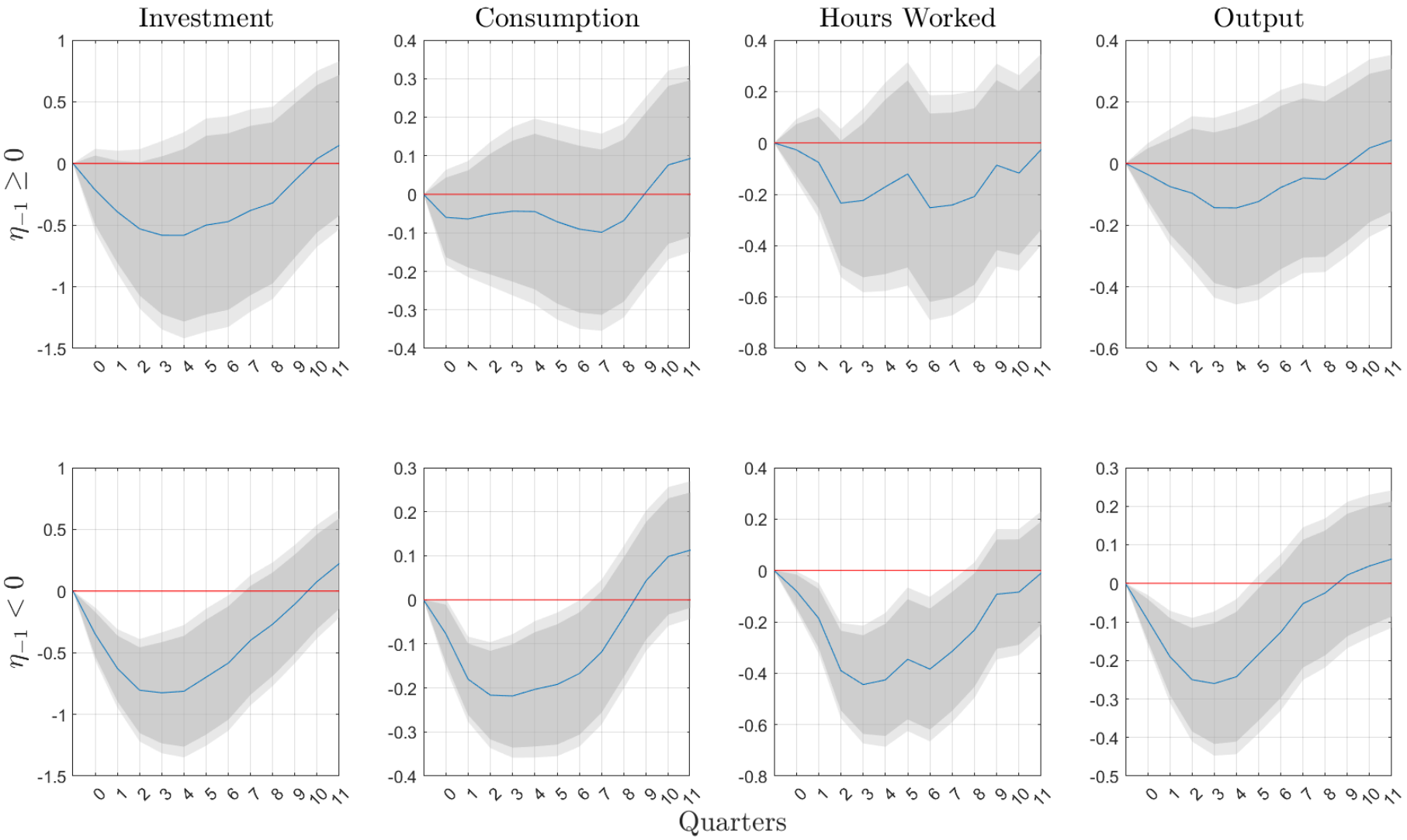

Figure: Dark grey: 90% confidence bands. Light grey: 95% confidence bands. Y-axis in percentage points. Monthly frequency LMN

Financial Variables: shock to VXO

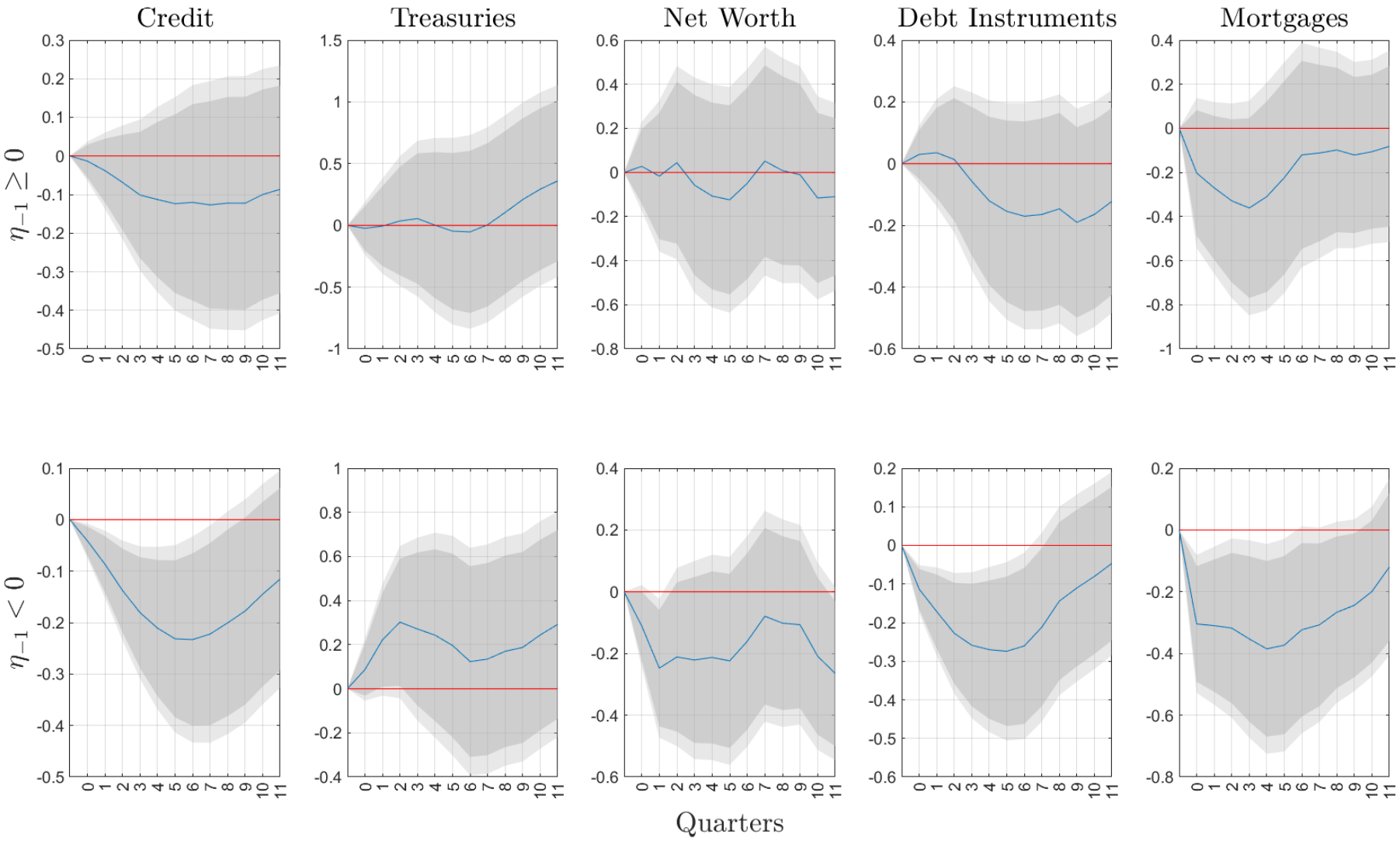

Figure: Dark grey: 90% confidence bands. Light grey: 95% confidence bands. Y-axis in percentage points. LMN Debt Instruments

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Setup

- Real Business Cycle model with financial sector *à la* Gertler and Karadi (2011)
  - HH cannot directly hold capital and saves through deposits
  - Financial frictions: intermediaries can abscond portion of assets
- Macro-uncertainty: TFP
- **Financial uncertainty**: random quantity of deposits is diverted from (to) the financial sector to (from) gvt bonds
  - Convenient way to shock the tightness of financial constraint
  - Real life banks rely on stable deposit + volatile funding Chart
  - Interpretation: HH delegates portfolio choice to a noise trader
- Global solution to construct state dependent impulse responses.

Household

$$\max_{C_t, A_t, L_t} \mathbb{E}_0 \left[\sum_{t \geq 0} \beta^t \frac{(C_t - \chi L_t)^{1-\gamma}}{1-\gamma} \right]$$

$$\text{s.t. } C_t + D_t = W_t L_t + R_{t-1} D_{t-1} + T_t$$

- Euler Equation

$$\mathbb{E}_t[\Lambda_{t,t+1}] R_t = 1$$

$$\Lambda_{t,t+1} = \beta \left(\frac{C_{t+1} - \chi L_{t+1}}{C_t - \chi L_t} \right)^{-\gamma}$$

- Labor supply

$$W_t = -\frac{\partial U_t / \partial L_t}{\partial U_t / \partial C_t} (= \chi)$$

Intermediaries

$$V_t = \max_{N_{t+1}, K_{t+1}} \mathbb{E}_t \left\{ \Lambda_{t,t+1} [(1-\sigma) N_{t+1} + \sigma V_{t+1}] \right\}$$

s.t.

$$Q_t K_{t+1} = N_t + \widetilde{D}_t \qquad \text{Flow of Funds}$$

$$N_{t+1} = R^k_{t+1} Q_t K_{t+1} - R_t \widetilde{D}_t \qquad \text{Net worth LOM}$$

$$V_t \geq \theta Q_t K_{t+1} \qquad \text{Financial constraint}$$

$$\widetilde{D}_t = e^{\sigma_t^\nu \nu_t} D_t \qquad \nu_t \overset{iid}{\sim} \mathcal{N}(0,1)$$

$$\sigma_t^\nu = (1-\rho_\sigma)\overline{\sigma^\nu} + \rho_\sigma \sigma_{t-1}^\nu + \varepsilon_t^\sigma \qquad \textbf{financial uncertainty}$$

Optimality condition:

$$\mathbb{E}_t [\Lambda_{t,t+1} (1 - \sigma + \sigma \underbrace{\psi_{t+1}}_{\frac{\partial V_{t+1}}{\partial N_{t+1}}}) (R^k_{t+1} - R_t)] = \mu_t \theta.$$

Production and Government

- Final good producer

$$\max_{L_t} e^{Z_t} K_t^{\alpha} L_t^{1-\alpha} - W_t L_t - Q_t I_t$$

$$\text{s.t. } W_t L_t \leq \zeta Q_t K_t$$

$$Z_t = (1-\rho_z)\bar{z} + \rho_z Z_{t-1} + \sigma^z \varepsilon_t^z$$

- K_t producers

$$\max_{I_t} Q_t I_t - \Phi\left(\frac{I_t}{K_t}\right) K_t$$

$$Q_t = \Phi'\left(\frac{I_t}{K_t}\right)$$

- Capital LOM

$$K_{t+1} = (1-\delta)K_t + I_t$$

- Government budget constraint

$$B_t + T_t = R_{t-1} B_{t-1}$$

Equilibrium

- Labor mkt equilibrium: fraction f of firms is assumed constrained

$$\chi = f\frac{\zeta Q_t K_t}{L_t} + (1-f)(1-\alpha)e^{Z_t}K_t^{\alpha}L_t^{-\alpha}$$

- Financial Markets equilibrium:

$$N_{t+1} = \sigma(R_{t+1}^k Q_t K_{t+1} - R_t D_t) + \omega Q_t K_{t+1}$$
$$D_t = B_t + \widetilde{D}_t$$

- Goods mkt equilibrium

$$C_t + \Phi\left(\frac{I_t}{K_t}\right)K_t = Y_t$$

Calibration

Variable	Parameter	Value	Source
RBC			**Standard**
Discount factor	β	0.995	
Risk aversion	γ	1	
Capital share	α	0.33	
Depreciation	δ	0.025	
Mean of TFP	$\bar{z}$	0	
Persistence of TFP	ρ_z	0.95	
S.d. of TFP innovations	σ_z	0.01	
Financial Sector			**GK (2011)**
Surviving bankers	σ	0.972	
Transfer to new bankers	ω	0.002	
Fraction of divertible assets	θ	0.381	
Calibrated Parameters			**Target Statistic**
Disutility of labor	χ	0.4352	$\mathrm{g}Y/\mathrm{g}L = 2.21$
Fraction of constrained firms	f	0.4233	$Var(\mathrm{g}Q)/Var(\mathrm{g}L) = 2.23$
Fraction of pledgeable assets	ζ	0.0933	$\mathrm{g}Q/\mathrm{g}L = 2.43$
Mean of σ_t^ν (s.d. of ν_t)	$\overline{\sigma^\nu}$	0.0179	$\frac{s.d.}{avg}$ g.r. bank funding
Persistence of σ_t^ν	ρ_σ	0.9	IRF persistence $R^{exc}_{SP500} = 0.98$

Solution Algorithm Calibration Details

Model Implied Conditional IRF

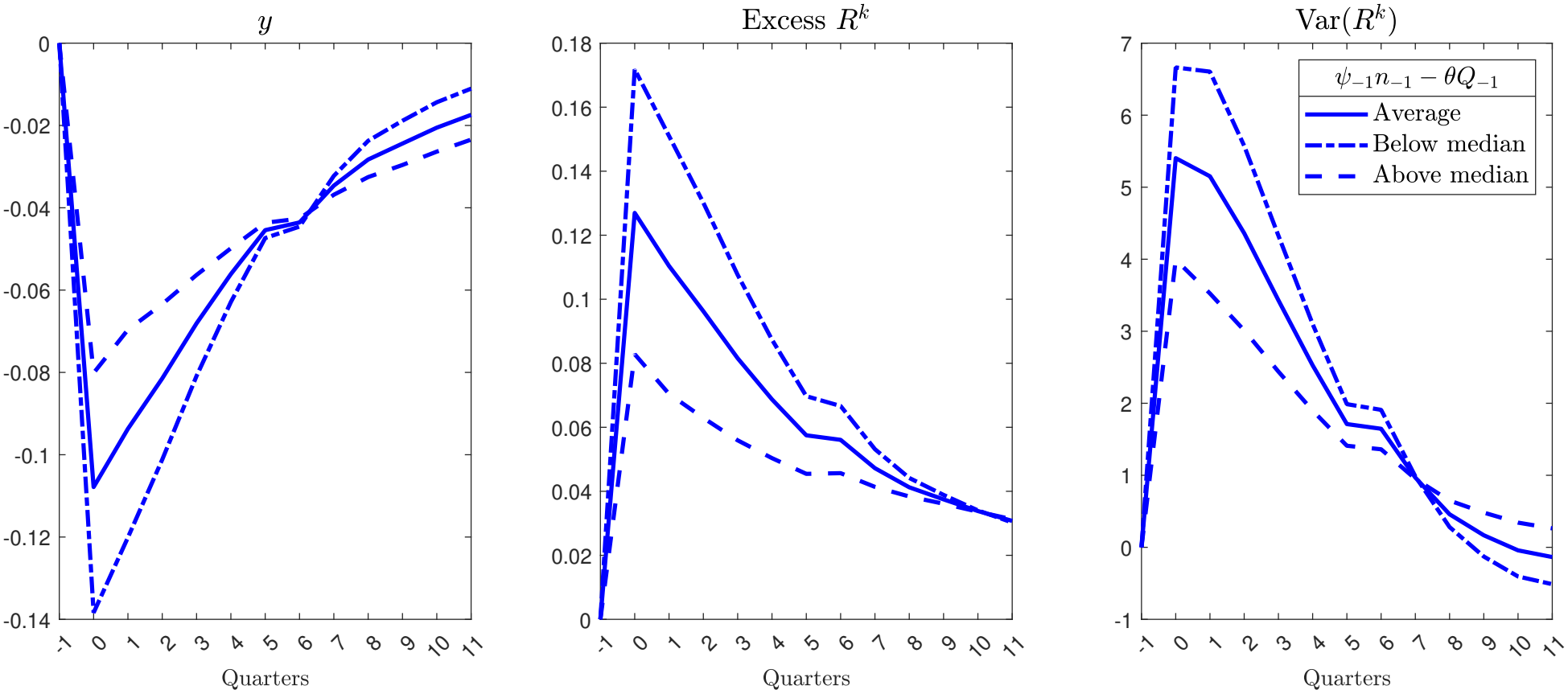

Figure: IRF to a 2× shock to σ_t^ν, conditional on the initial state of the economy. Quantities are computed as a fraction of the stock of capital: $x_t = \frac{X_t}{K_t}$. Other variables

Empirical vs Theoretical IRF

Table: $IRF(h \mid \text{'distressed'})/IRF(h \mid \text{'not distressed'})$ in the model and in the data for selected variables. Plots

h	i		y		$\mathbb{E}[R^k] - R$	
	Model	Data	Model	Data	Model	Data
1	1.63	1.19	1.65	1.71	2.39	3.92
2	1.65	1.44	1.67	1.32	2.51	5.26
3	1.54	1.36	1.55	1.26	2.42	3.98
4	1.41	1.6	1.42	1.6	2.27	4.37

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- Financial sector balance sheet explains magnitude of excess returns countercyclicality triggered by uncertainty shocks
- Novel source of financial uncertainty in macro-finance RBC generates the differential responses seen in the data
- Avenues for future research:
  - Introduce nominal frictions
  - Micro-found noise traders
  - Policy response.

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Appendix

Literature Review

- **On excess return cyclicality**
  - Fama and French (1989), Ferson and Harvey (1991), Lettau and Ludvigson (2001), Lettau and Ludvigson (2009), Backus et al. (2010), Lustig and Verdelhan (2012), Nagel and Xu (2023) for stocks;
  - Gilchrist and Zakrajšek (2012) for corporate bonds;
  - Cochrane and Piazzesi (2005) and Ludvigson and Ng (2009) for U.S. gvt bonds;
  - Lettau et al. (2014) for commodities, sovereign bonds and currencies.
- **On uncertainty shocks**
  - Bloom (2009), Basu and Bundick (2017), Bloom et al. (2018), Fernandez-Villaverde et al. (2011), Fernandez-Villaverde et al. (2015), Born and Pfeifer (2014) without credit constraints;
  - Christiano et al. (2014), Cesa-Bianchi and Fernandez-Corugedo (2018), Gilchrist et al. (2014), Fernandez-Villaverde and Guerrón-Quintana (2020), Chatterjee et al. (2020), Fang and Liu (2021) with credit constraints.
- **On intermediary-based asset pricing**
  - He and Krishnamurthy (2013), Brunnermeier and Sannikov (2014), He et al. (2017) in finance;
  - Bernanke et al. (1999) Gertler and Karadi (2011), Gertler et al. (2020) in macro.

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Bloom (2009)

- Construct volatility index by splicing VXO with volatility of SP500 index
- Run a VAR with

$$\mathbf{y}_t = [R_t, vola_t, ffr_t, cpi_t, wages_t, empl_t, hours_t, ip_t]$$

- Shock identification: ordering and Choleski decomposition
- $vola_t$ shock contemporaneously affects the stock market, then prices and finally quantitites.

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Jurado Ludvigson Ng (2015) and Ludvigson Ma Ng (2021)

- 132 macro variables divided in 8 groups: income and production, labor mkt, housing, consumption and investment, money and credit, interest and exchange rates, prices;
- 147 financial variables: dividends, prices, d-p ratios, common factors and returns on a large number of portfolios;
- Estimate a FAVAR model for all the variables
- compute $h-$step ahead forecast and forecast errors;
- Uncertainty at horizon h = st. dev. of forecast errors at horizon h
- Shock identification through VAR $[U_{Mt}, ip_t, U_{Ft}]$ and "shock-based restrictions" (event constraints).

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Data and Variables I

- "Equity-invcap" variable on COMPUSTAT for all the firms in the "Financial" sector according to the GICS classification
- **Excess Returns**
  - SP500 returns from CRSP
  - Excess returns computed as the monthly 10yr return on stock minus the monthly 10yr risk free rate taken from Kenneth French website
  - GZ spread and EBP from Gilchrist and Zakrajšek (2012)
  - Term spread = 10 years-3 months Treasury rate at constant maturity. Source: FRED
  - Mortgage spread = 30 years fixed rate mortgage - 10 years Treasury rate. Source: FRED
- **Balance sheet items**
  - Credit to nonfinancial businesses by financial institutions (depository institutions and others). Source: BIS statistics. Credit includes both loans and debt securities
  - "Domestic Financial Sectors; Treasury Securities; Asset, Level", Source: FRED
  - "Domestic Financial Sector; Total Financial Asset, Level" and "Total Financial Liabilities" to compute net worth. Soruce: FRED
  - "All Sectors; Total Mortgages; Asset, Level". Source: FRED

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Data and Variables II

- Controls for excess returns: $ip_t, UNC_t, \log Q_t, \pi_t$ (FRED), common factors (Kenneth French website)
- Controls for macro aggregates: $ip_t, FFR_t, 10Y - 3m, \pi_t, \log Q_t$ (FRED)
- **Debt Instruments** (see Gorton et al. (2012))
  - "Money Market Funds; Total Financial Assets"
  - "Domestic Financial Sectors; Checkable Deposits and Currency"
  - "Domestic Financial Sectors; Federal Funds and Security Repurchase Agreements"
  - "Private Depository Institutions; Total Time and Savings Deposits"
  - "Finance Companies; Commercial Paper"
  - "Private Depository Institutions; Bankers' Acceptances"
  - "Domestic Financial Sectors; Total Miscellaneous Liabilities"
- Bank funding data from Bowman, Scotti, and Vojtech (2020), who use Call Report and FR-Y9C.

Specification IRFs

Uncertainty indexes

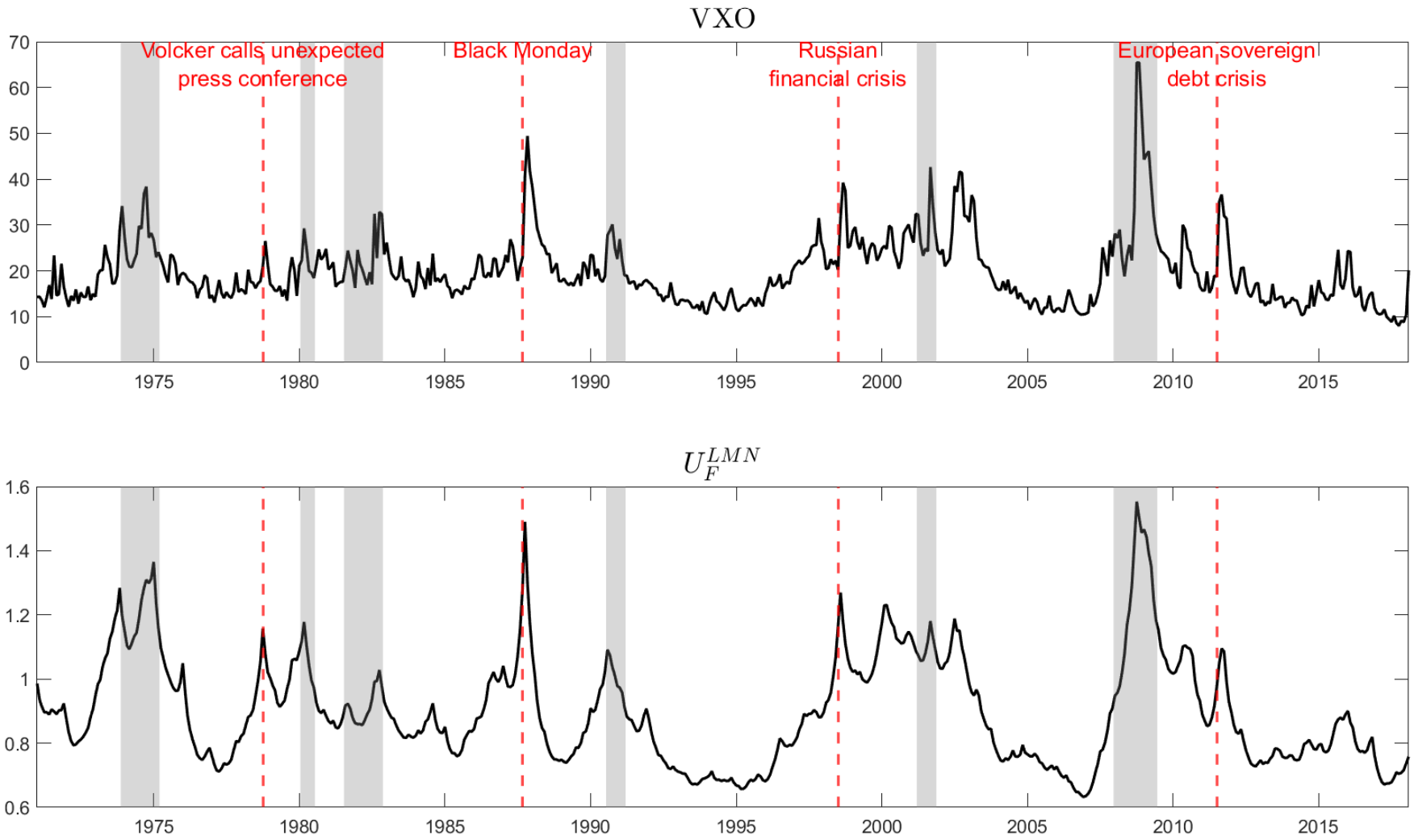


Figure: VXO and U_F^{LMN} indexes with significant events in financial markets history. Back

Uncertainty shocks

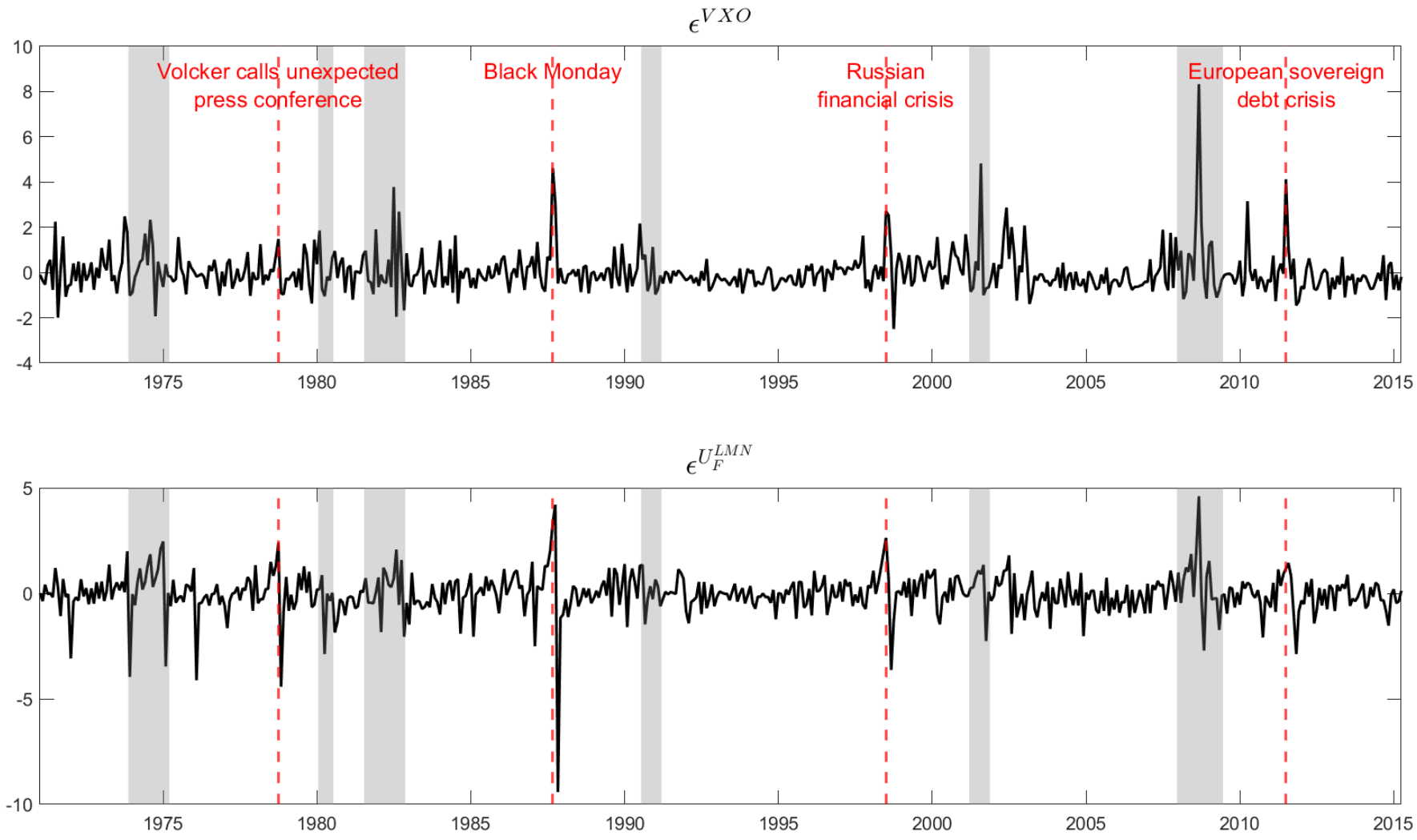


Figure: Identified structural shocks to VXO and U_F^{LMN} with significant events in financial markets history. Back

Excess Returns: shock to U_F^{LMN}

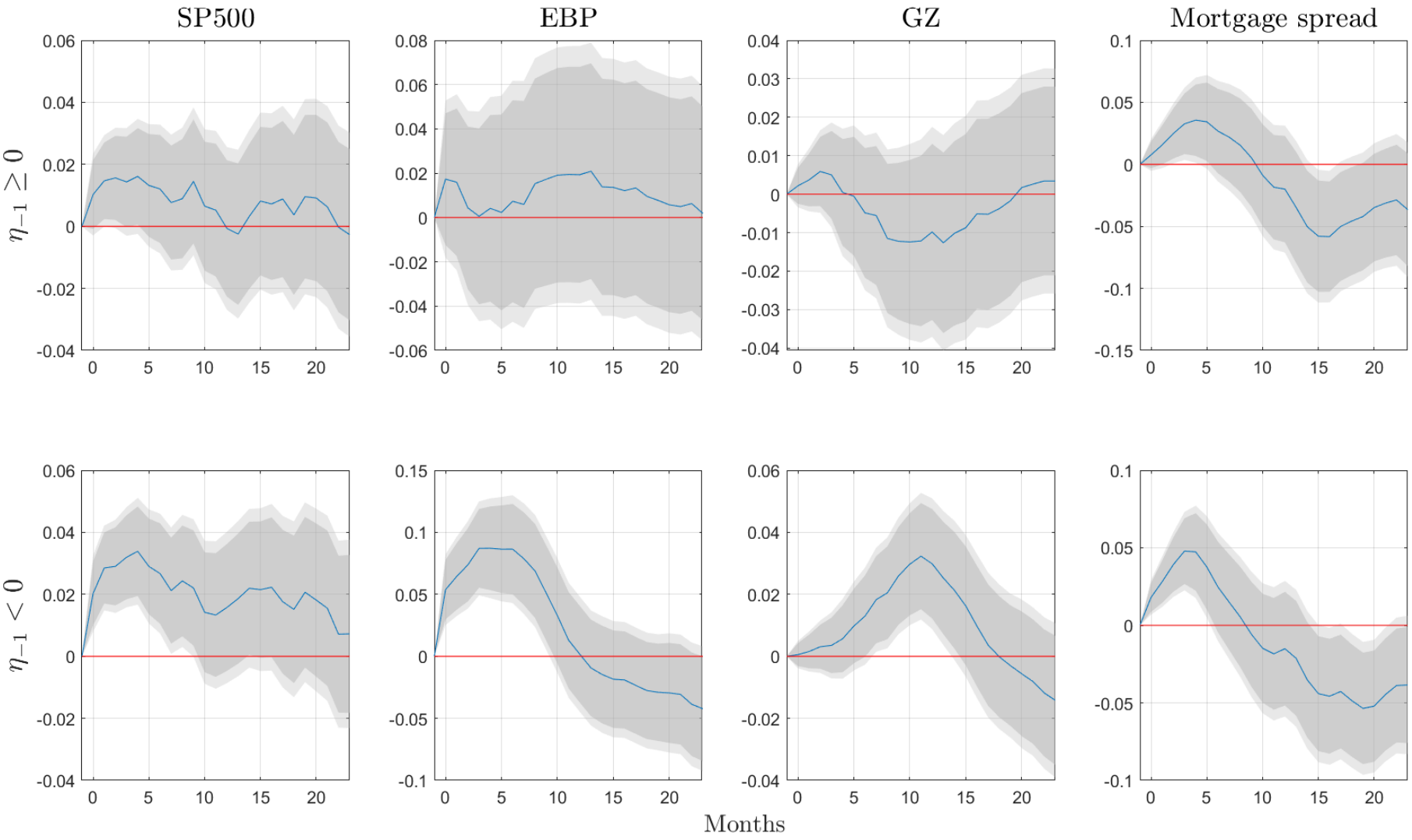
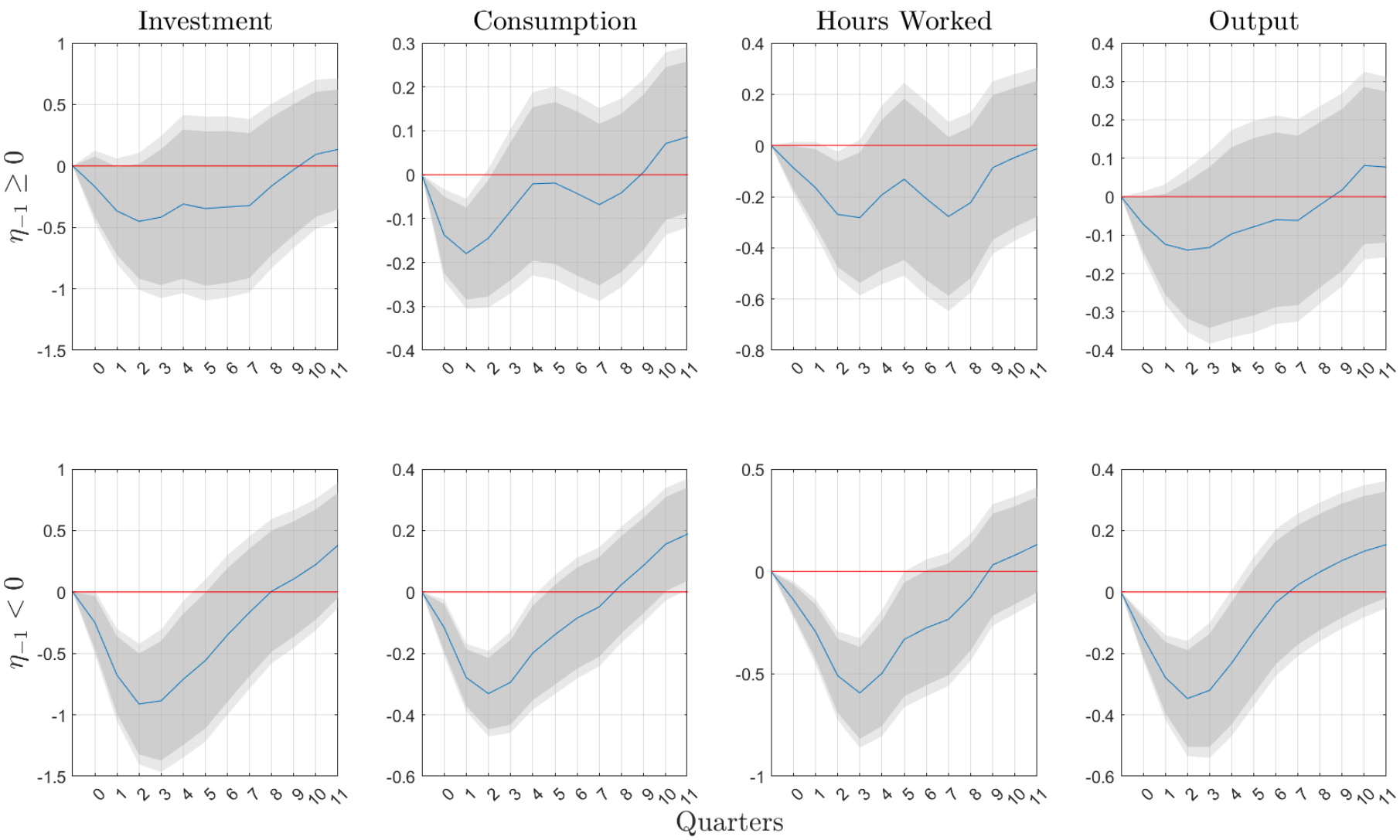


Figure: Dark grey: 90% confidence bands. Light grey: 95% confidence bands. Y-axis in percentage points. Back

Real Variables: shock to U_F^{LMN}



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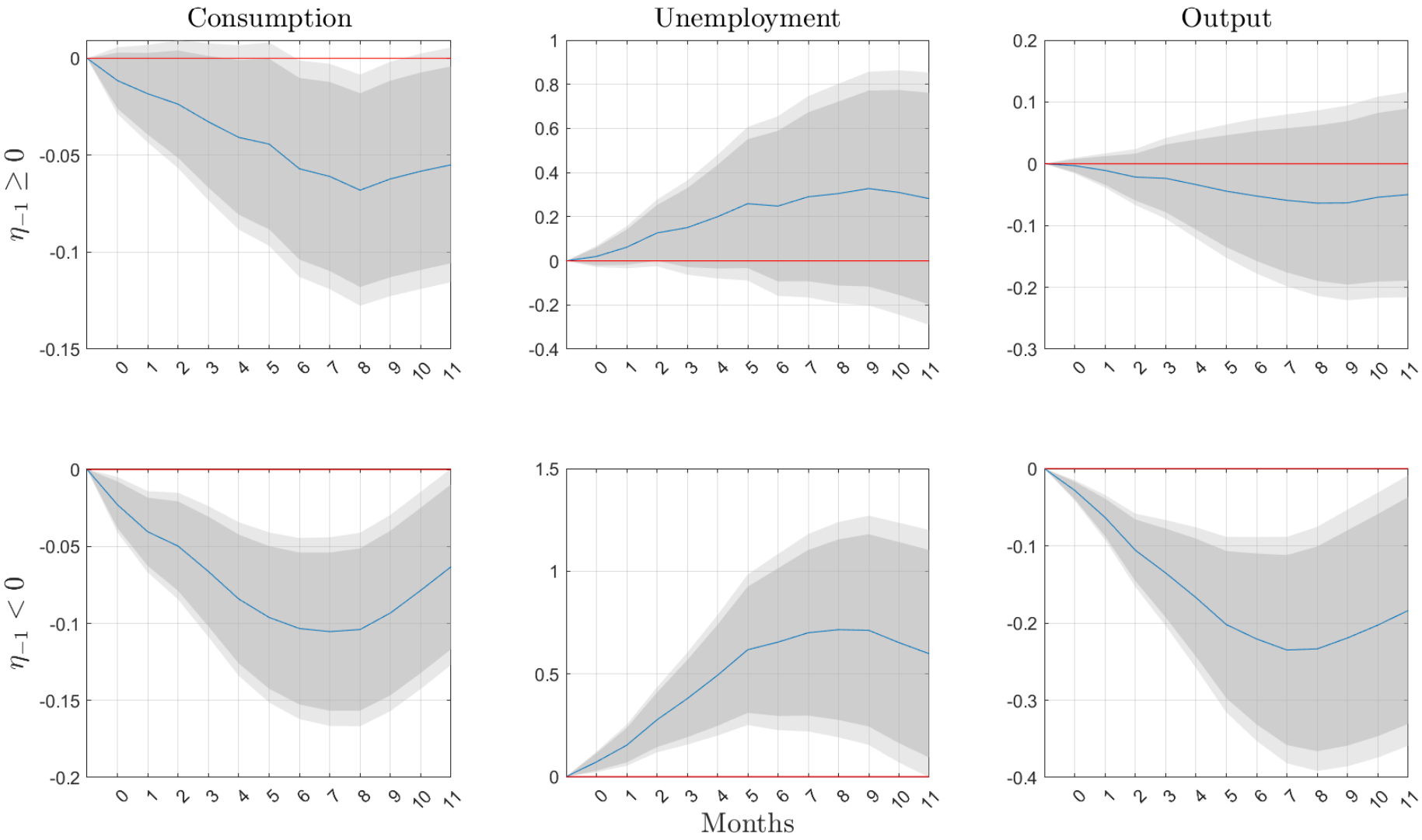
Real Variables: shock to U_F^{LMN}



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Financial Variables: shock to U_F^{LMN}

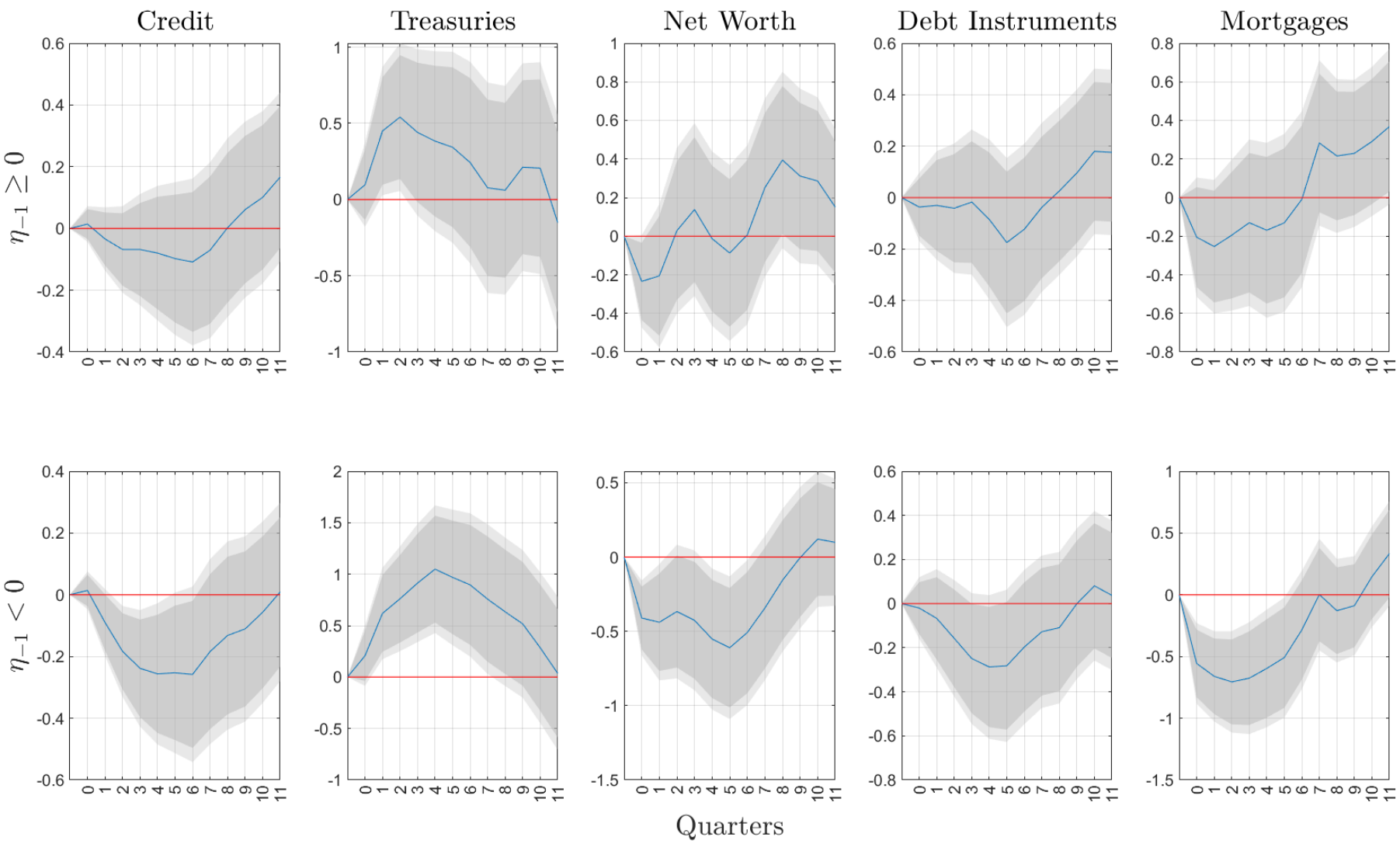

Figure: Dark grey: 90% confidence bands. Light grey: 95% confidence bands. Y-axis in percentage points. Back

Solution specification I

- Normalize quantity variables by K and denote $x = \frac{X}{K}$
- States: $S = (\log p, Z)$ with $p_{t-1} = R_{t-1} A_{t-1}$
- Eq. Functions $\Theta = (c, \ell, u, i, y, R, \psi, Q, n, p')$, where $u = c - \chi \ell$
- (ℓ, R, ψ) pin down (c, u, i, y, Q, n, p'). Static Equations

Solution specification II

(ℓ, R, ψ) come from the solution of the system:

$$\beta u^{\gamma}(1-\delta+i)^{-\gamma}\mathbb{E}\left[(u')^{-\gamma}(1-\sigma+\sigma\psi')((R^k)'-R)\right]=\theta\mu \quad (1)$$

$$\beta u^{\gamma}(1-\delta+i)^{-\gamma}\mathbb{E}\left[(u')^{-\gamma}\right]R=1 \quad (2)$$

$$\psi=\beta u^{\gamma}(1-\delta+i)^{-\gamma}\mathbb{E}\left[(u')^{-\gamma}(1-\sigma+\sigma\psi')\frac{n'}{n}\right] \quad (3)$$

$$\mu(\psi n-\theta Q)\geq 0 \quad (4)$$

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Solution specification III

Rewrite Equation (3) and condition (4) as

$$\mu = \max\left\{1 - \beta\frac{u^{\gamma}(1-\delta+i)^{-\gamma}\mathbb{E}\left[(u')^{-\gamma}(1-\sigma+\sigma\psi')\right]Rn}{\theta Q}, 0\right\} \tag{5}$$

$$\psi = \max\left\{\beta u^{\gamma}(1-\delta+i)^{-\gamma}\mathbb{E}\left[(u')^{-\gamma}(1-\sigma+\sigma\psi')\right]R, \frac{\theta Q}{n'}\right\} \tag{6}$$

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Algorithm

- Start with small value of θ: solution shouldn't be too far from unconstrained (RBC) solution
- Initial guess ℓ_0: RBC solution; R_0 derived from (2) and $\psi_0 = \max\left\{\frac{\theta Q_0}{n_0'}, 1\right\}$
- Given (ℓ_n, R_n, ψ_n) update ℓ by solving for (1) and (5);
- Given $(\ell_{n+1}, R_n, \psi_n)$, update R using (2) and ψ using (6);

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Model Static Equations

- $Q = \frac{\chi\ell - (1-f)(1-\alpha)e^{Z}\ell^{1-\alpha}}{f\zeta}$
- $y = e^{Z}\ell^{1-\alpha}$
- $i = (\Phi')^{-1}(Q) = a_1^{-\frac{1}{\xi}} Q^{\frac{1}{\xi}}$
- $c = y - \Phi(i)$
- $n'(1-\delta+i) = \sigma\left\{[\alpha y + (1-\delta)Q] - p\right\} + \omega Q$
- $p' = R(Q - n')$

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Calibration Details

Table: Target statistics for parameter calibration. Shaded parameters are determined jointly.

Parameter	Target statistic	Model	Data
f	$\frac{Var(Q_t K_t)}{Var(L_t)}$	2	2.23
ζ	$\frac{QK}{L}$	2.52	2.43
χ	$\frac{K}{L}$	1.34	1.4
$\overline{\sigma^{\nu}}$	$\frac{s.d.}{avg}$ g.r. of total bank funding	0.0179	0.0179
ρ_{σ}	IRF persistence for R^{exc}_{SP500}	0.9023	0.988

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Bank funding

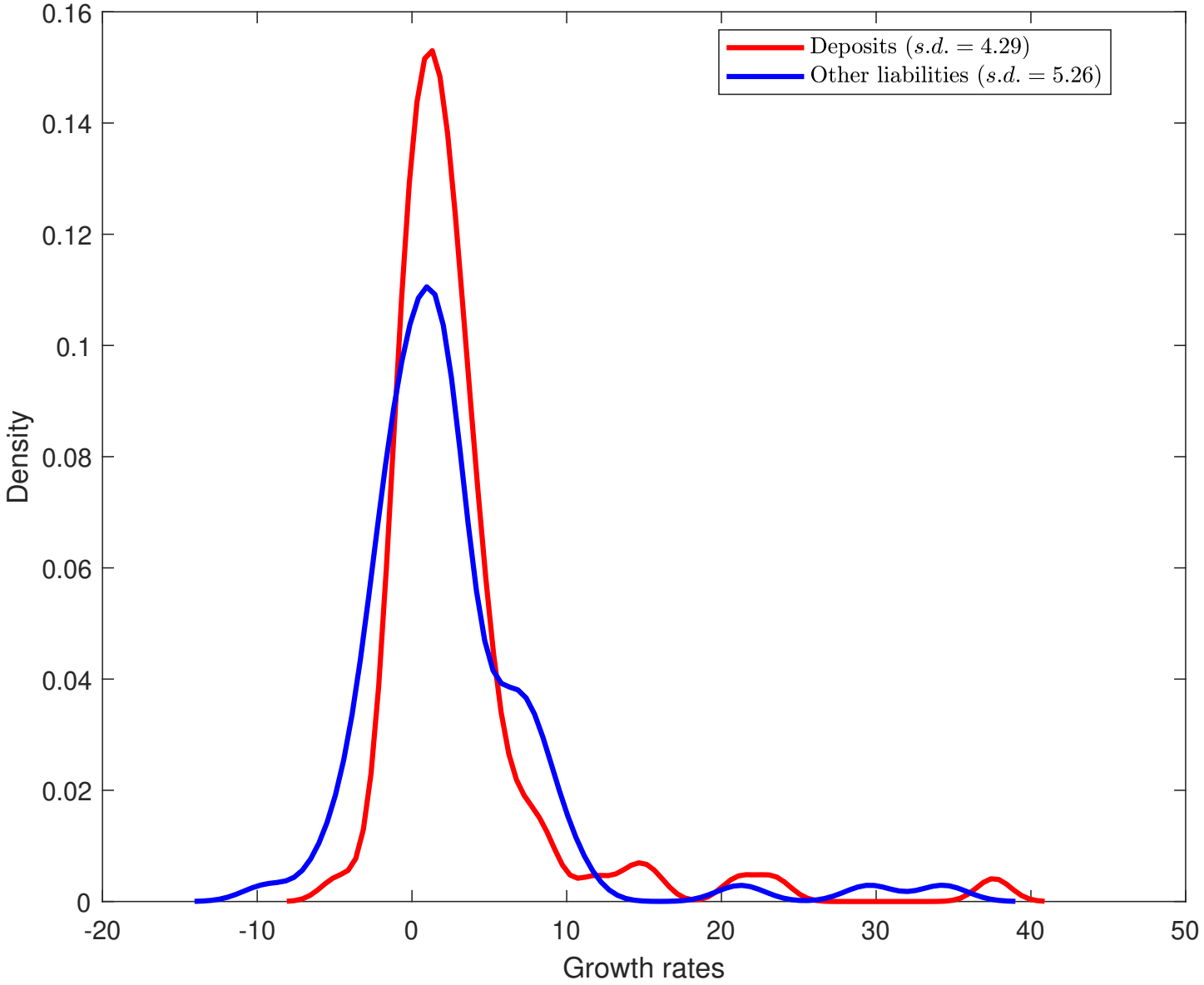

Figure: Kernel density of growth rates of bank funding by type. Data source: Bowman, Scotti, and Vojtech (2020), FR-Y9C, Call Report. Back

Dynamics of a shock ν_t

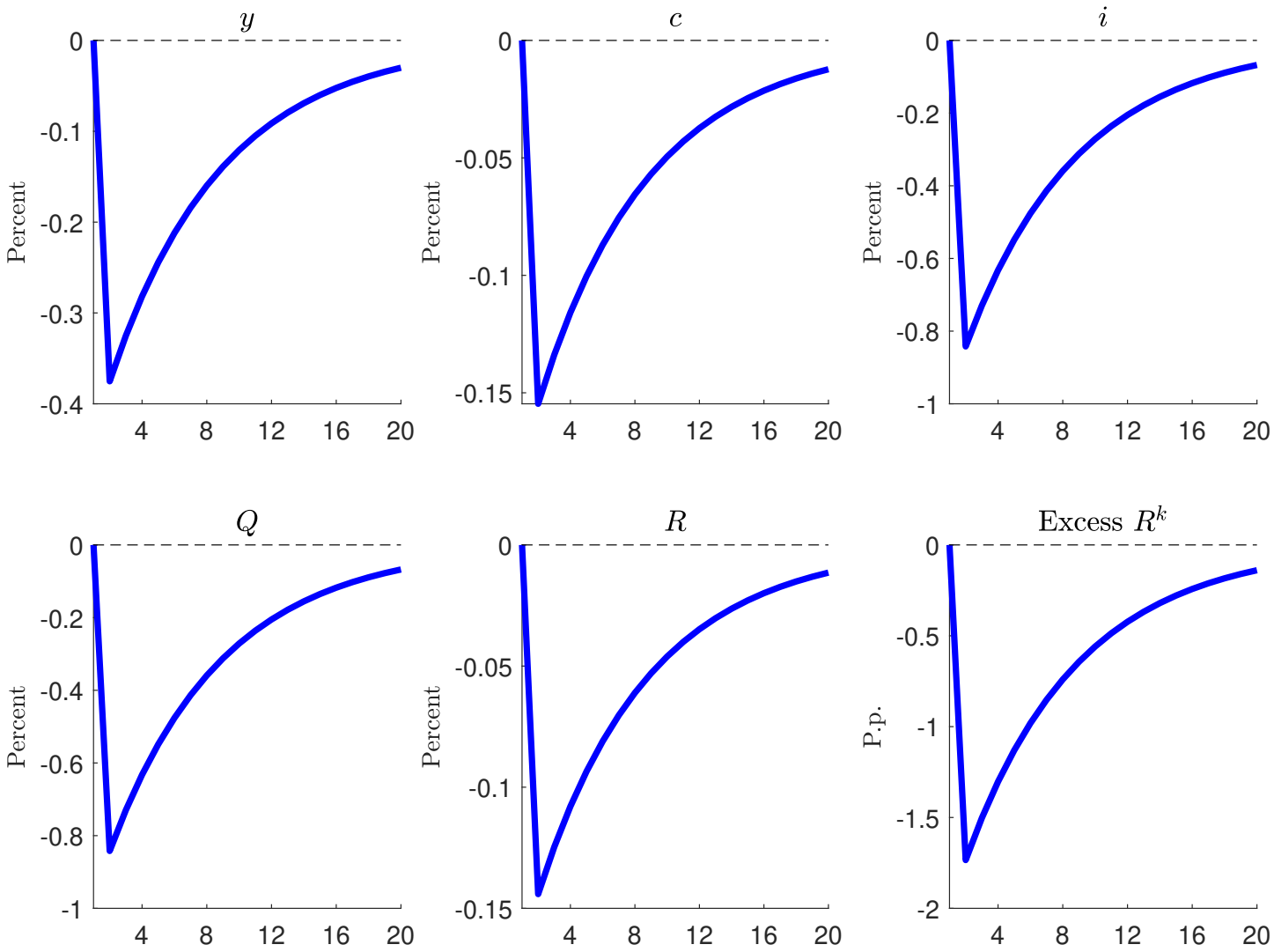

Figure: IRF to 1 s.d. shock to ν_t. Back

Real Variables: shock to VXO

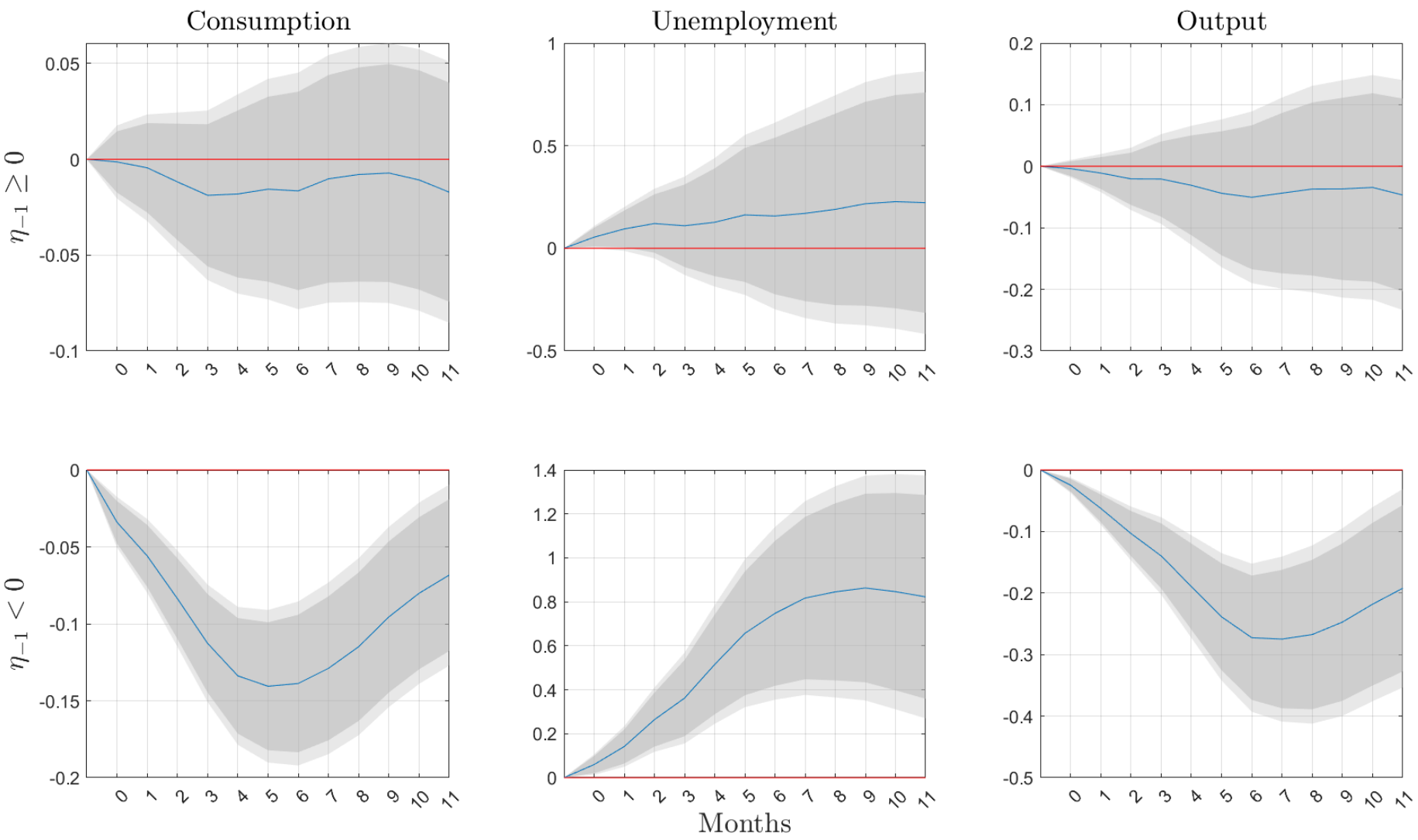

Figure: Dark grey: 90 confidence bands. Light grey: 95 confidence bands. Y-axis in percentage points. Back

Summary of η

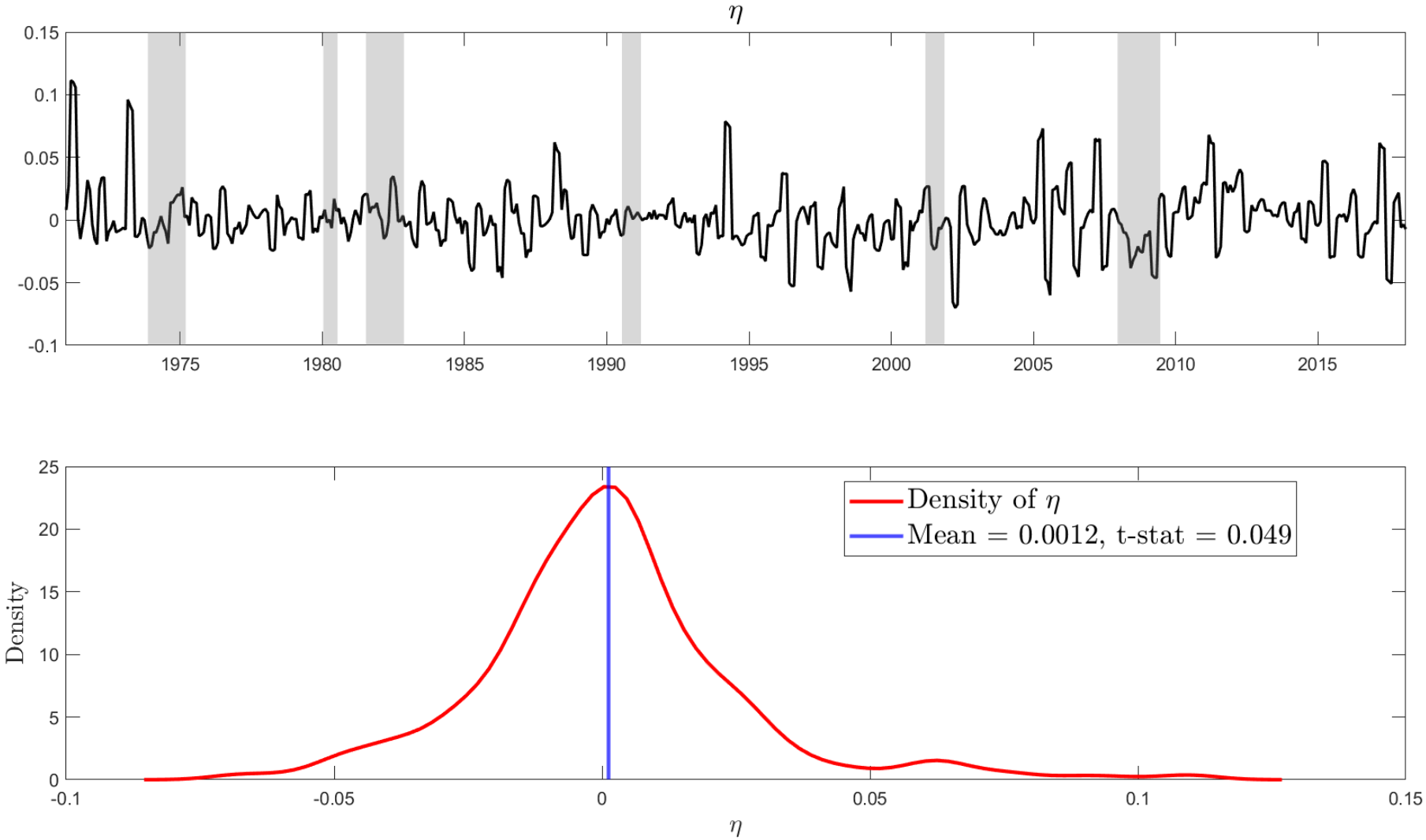

Figure: Time series and kernel density of η. Back

Model Implied Conditional IRF

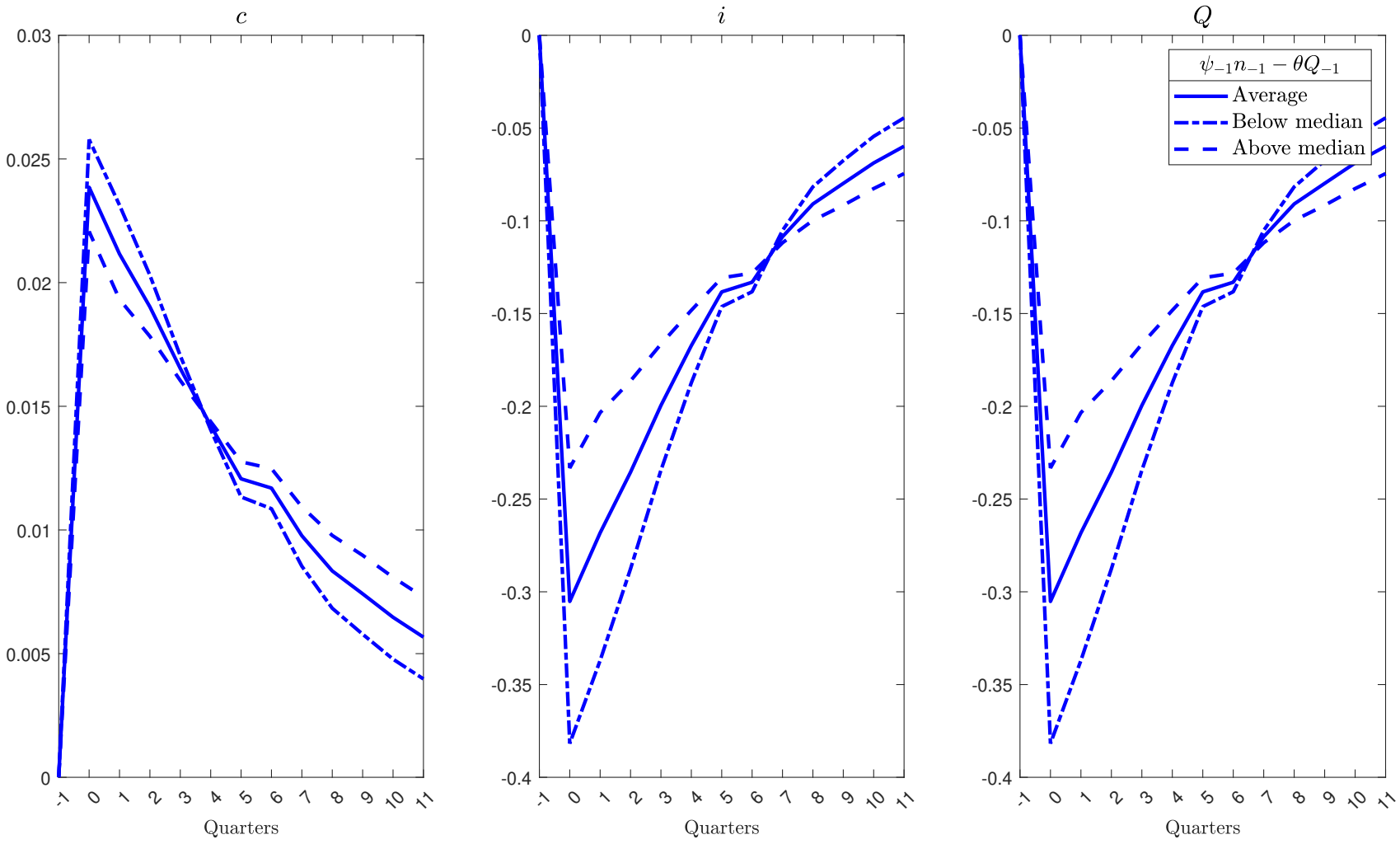

Figure: IRF to a 2× shock to σ_t^{ν}, conditional on the initial state of the economy. Quantities are computed as a fraction of the stock of capital: $x_t = \frac{X_t}{K_t}$. Back

Empirical vs Theoretical IRF

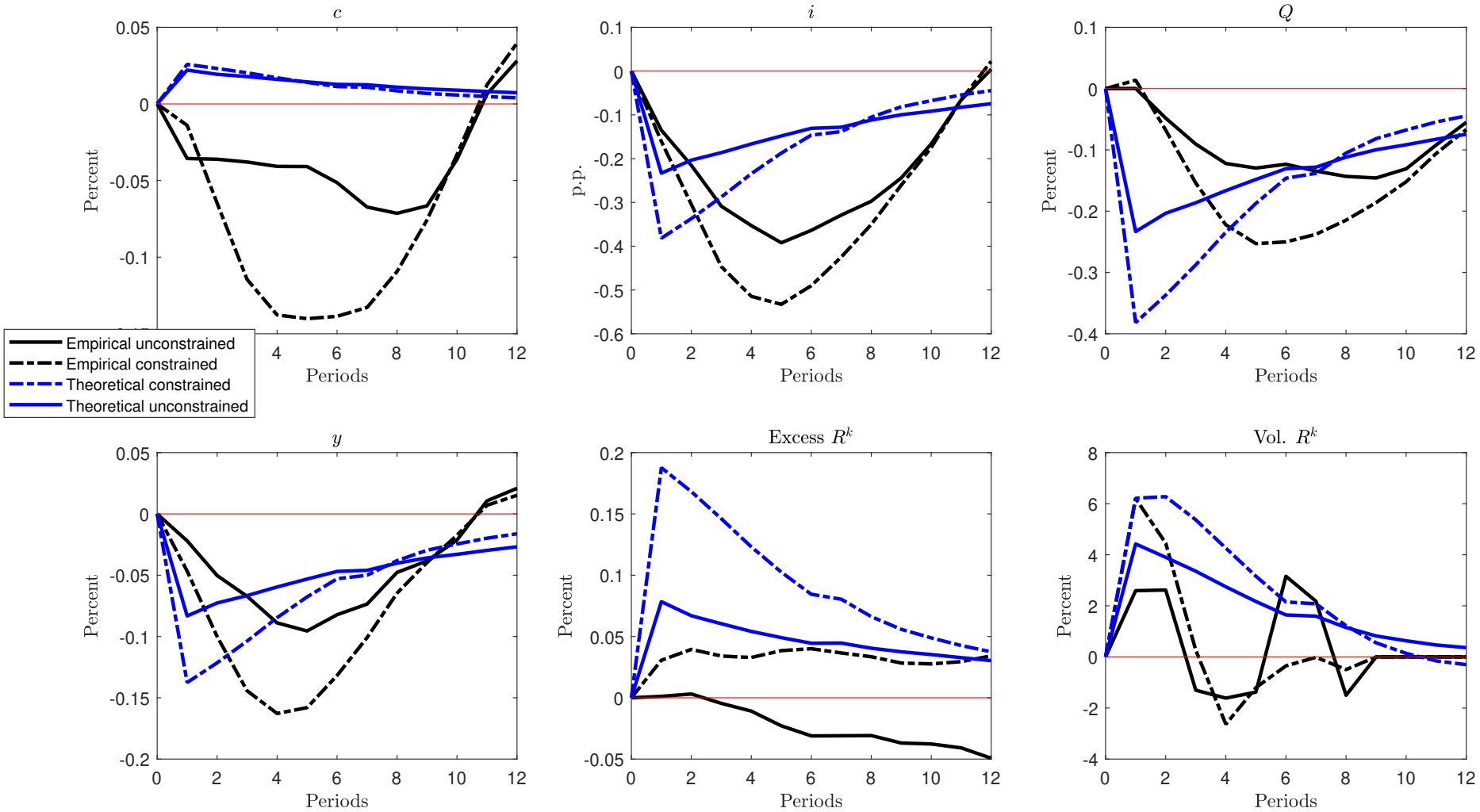

Figure: Model implied and estimated IRF to a 2× shock to σ_t^{ν}, conditional on the initial state of the economy. Quantities are computed as a fraction of the stock of capital: $x_t = \frac{X_t}{K_t}$. Back